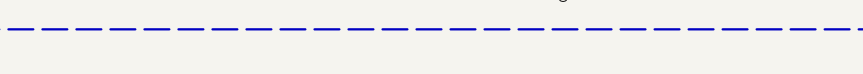
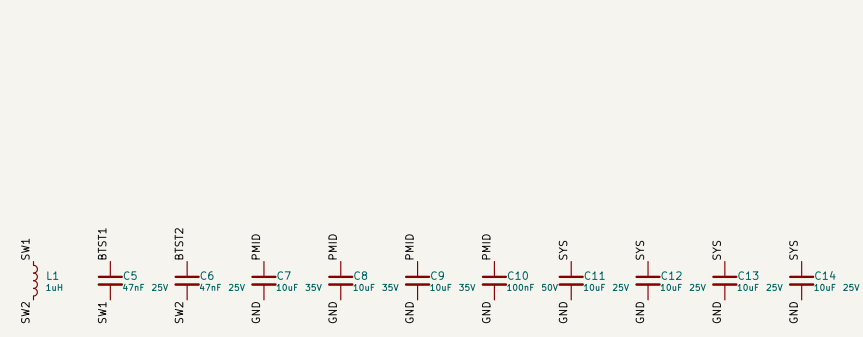


CC1/CC2 direct to CH224K (internal Rd). DP/DM of CH224K shorted together and NOT connected to the connector (WCH 'PD-only' mode) so USB D+/- go to the MCU. CFG2/CFG3 float = single-resistor mode. CFG1 56k = 15V. PG low = contract OK. CH224K VBUS sense pin left NC (allowed in PD-only mode; pin is only 13.5V rated).

3V3 = OR of: VBUS (charger OR a phone in the USB-C port) | battery via Q2 when the MCU asserts PWR_HOLD (lets the MCU finish balancing, put BQ25798 in shutdown + BQ76920 in SHIP after unplg, then release itself). Nothing else can power the MCU from the pack. Idle battery drain ~1-2uA total.

PROG 10.5k -> 3S / 1.5MHZ: POR defaults VREG=12.6V, ICHG=1A, VSYSMIN=9V.
No ACFTETS: VAC1/VAC2 tied to VBUS, ADCR1/2 to GND. IILM_HIZ->REGN (limit set by I2C)
/CE held HIGH by 100k -> No charging until MCU drives CHG_EN HIGH (after cell check).
BQ25798 D+/D- left open: MCU must clear AUTO_INDET_EN and set INDPM over I2C.
TS: 5.23k/30.1k + 10k NTC = 0.60C window (datasheet 7.3.9.5). Shutdown mode = 0.5uA.



PF0/PF1 = I2C1 SDA/SCL (AF1). PA11/PA12 remapped onto pins 17/18 for USB (SYSCFG PA11_PA12_RMP).
 PA0 /INT charger, PA1 PD_PG, PA2 CHG_EN, PA3 ALERT, PA4 BMS boot, PA5 VBUS/11 ADC, PA6 LED, PA7 PWR_HOLD.
 PB1 = FAST_CHG_N (JP2 to GND, internal pull-up): open = slow charge (default), bridged = fast charge.
 Bridge JP1 + plug USB = ST ROM DFU bootloader. Backup: ST-Link on test pads TP5 SWDIO, TP6 SWCLK, TP7 3V3, TP8 GND.

3S per datasheet Table 9-2: VC3-VC4 = cell3, VC4=VC3=VC2 shorted, VC2-VC1 = cell2, VC1-VC0 = cell1.
Rc=56R / Cc=1uF cell filters -> -37mA internal balancing (datasheet min Rc 40R, max 50mA) (never balance adjacent cells together).
SRP/SRN to VSS (no coulomb counter), CHG/DSG unused. Sits in SHIP mode (-0.6uA) when idle.
Balance lead = SENSE ONLY (charging goes through the Deans main lead), so readings stay accurate.

